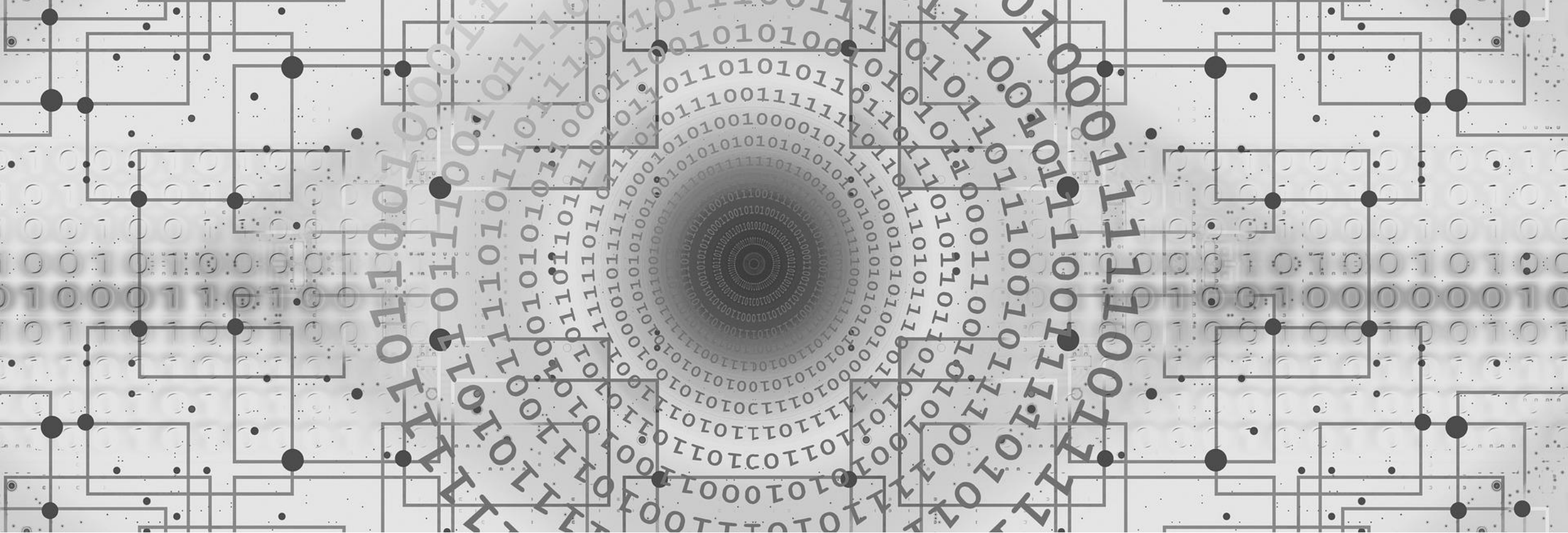




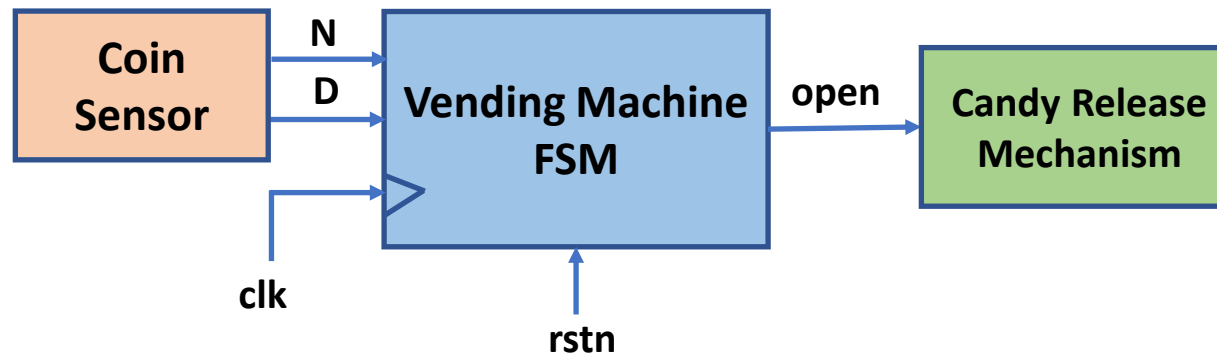
Homework-7d : Vending Machine FSM Minimization

ECE-111 Advanced Digital Design Project

Homework-7d : Vending Machine FSM Minimization

❑ Consider a Vending Machine with below mentioned operation :

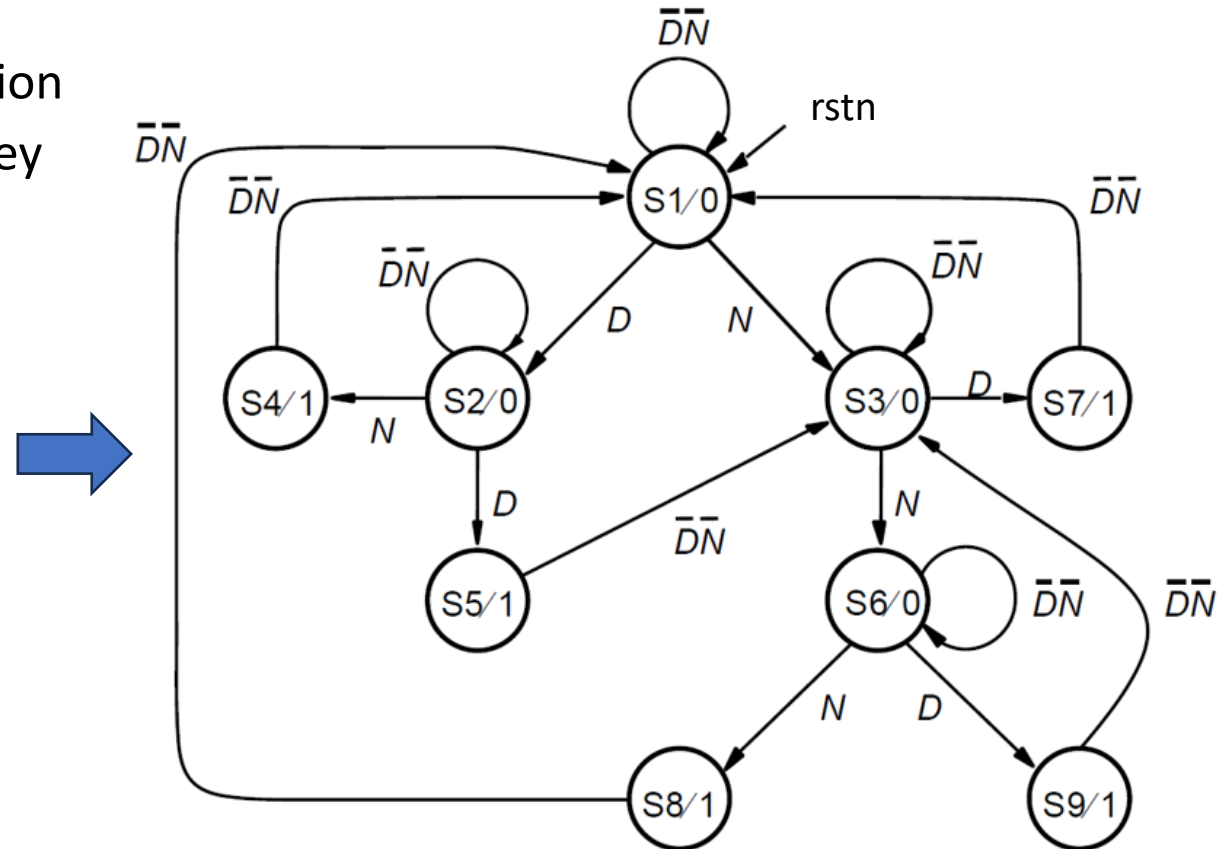
- Dispatches a pack of candy upon deposit of 15 cents or more by the customer
- Has single coin slot and only accepts nickels and dimes. Doesn't take pennies or quarters
- If 15 cents is deposited, then vending machine will dispatch the candy and then move to reset / IDLE state automatically
- **If 20 cents is deposited, machine will not return the change, it will first dispatch the candy then it will credit the buyer with 5 cents and wait for the buyer to make a second purchase**



Homework-7d : Vending Machine FSM Minimization

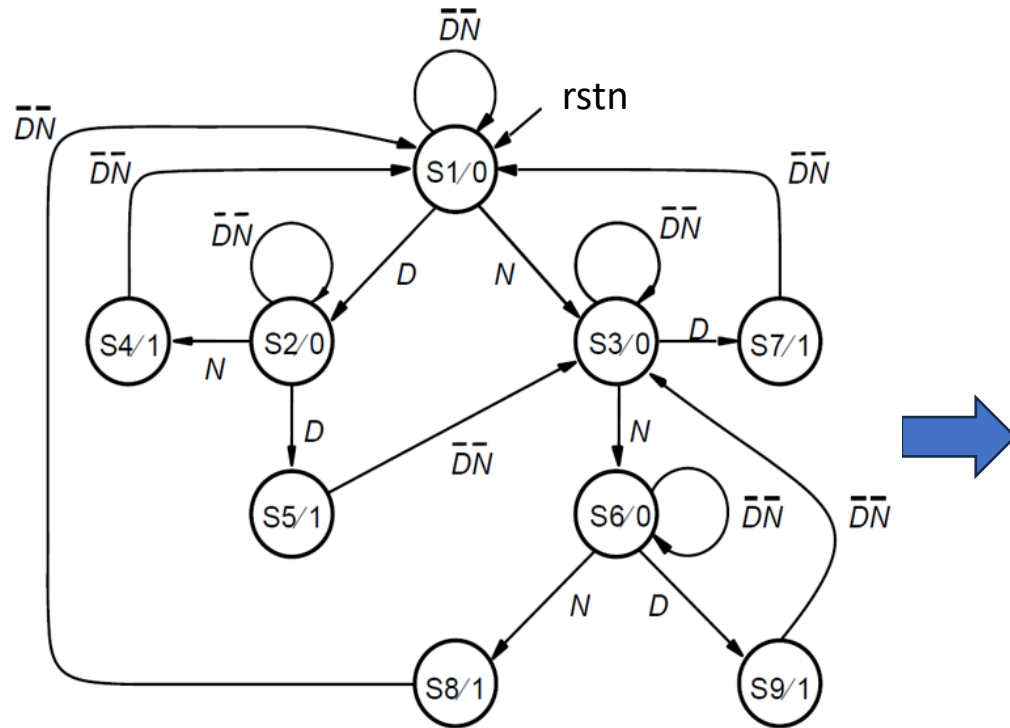
❑ Vending Machine **Moore** State Machine Diagram, before FSM minimization

- Consider two inputs : D (Dime) and N (Nickel)
- Total **9** states identified before FSM minimization
 - **S1** : Starting State or Idle State : No Money
 - **S2** : Dime deposited
 - **S3** : Nickel deposited
 - **S4** : S2 + Nickel deposited, and candy dispatched
 - **S5** : S2 + Dime deposited, and candy dispatched. **Extra 5 cents credit**
 - **S6** : S3 + Nickel deposited
 - **S7** : S3 + Dime deposited, and candy dispatched
 - **S8** : S6 + Nickel deposited, and candy dispatched
 - **S9** : S6 + Dime deposited, and candy dispatched. **Extra 5 cents credit**



Homework-7d : Vending Machine FSM Minimization

❑ Vending Machine State Transition Table, before FSM minimization

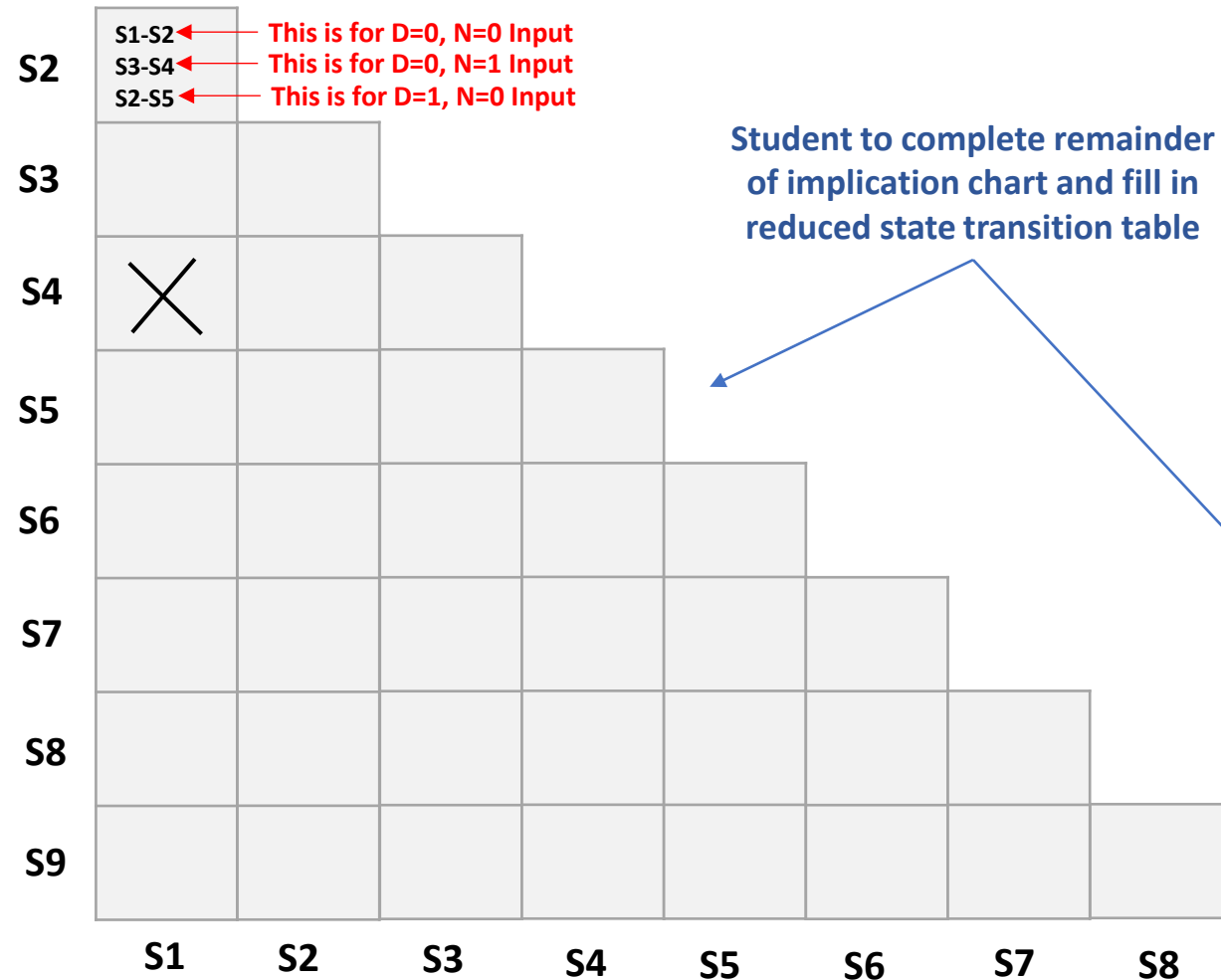


Present State	Next State				Output
	D=0, N=0	D=0, N=1	D=1, N=0	D=1, N=1	out
S1	S1	S3	S2	X	0
S2	S2	S4	S5	X	0
S3	S3	S6	S7	X	0
S4	S1	X	X	X	1
S5	S3	X	X	X	1
S6	S6	S8	S9	X	0
S7	S1	X	X	X	1
S8	S1	X	X	X	1
S9	S3	X	X	X	1

Homework-7d : Vending Machine FSM Minimization

□ Perform Vending Machine FSM minimization using implication chart technique

- **Step-1** : Construct implication chart
- **Step-2** : Derive reduced state transition table



State Transition Table before Minimization

Present State	Next State				Output
	D=0, N=0	D=0, N=1	D=1, N=0	D=1, N=1	out
S1	S1	S3	S2	X	0
S2	S2	S4	S5	X	0
S3	S3	S6	S7	X	0
S4	S1	X	X	X	1
S5	S3	X	X	X	1
S6	S6	S8	S9	X	0
S7	S1	X	X	X	1
S8	S1	X	X	X	1
S9	S3	X	X	X	1

Present State	Next State				Output
	D=0, N=0	D=0, N=1	D=1, N=0	D=1, N=1	out

State Transition Table after Optimization

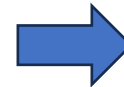
Note : Homework folder has a excel file with empty implication chart. Student may use the chart in excel to fill in for vending state machine. Student can take a snapshot of implication chart and provide it homework report

Homework-7d : Vending Machine FSM Minimization

- ❑ Create Vending Machine **Moore** state transition diagram after performing FSM minimization
 - Homework folder has a excel file with state transition empty table. Student may use the table in excel file to create the state transition table.
 - In homework report provide snapshot of the state transition table
 - Draw state transition table paper, take a picture and attach a snapshot of the transition table in homework report

State Transition Table after Optimization
(Student to fill in the table)

Present State	Next State				Output
	D=0, N=0	D=0, N=1	D=1, N=0	D=1, N=1	out



State Transition Diagram after FSM minimization
(Student to provide below)

Homework-7d : Vending Machine FSM Minimization

❑ Develop SystemVerilog code for reduced state Moore FSM for vending machine :

- Perform state minimization using implication chart technique
- Design state transition table and diagram for Moore implementation after performing FSM minimization
- Use binary encoding for state variables. Use either 2 always block or 1 always block FSM implementation
- Review Vending state transition table and Moore FSM state transition diagram for FSM code development
- Synthesize and review RTL netlist schematic, state machine viewer and resource usage
- Simulate Moore implementation using testbench provided and Review waveforms

❑ Report should include ?

- Snapshot of SystemVerilog FSM RTL code
- Snapshot of simulation waveform generated from Modelsim
 - Modelsim transcript window snapshot is not required since there are no prints from the testbench
- Snapshot of implication chart after it is constructed. Either handwritten or using template in excel provided
- Snapshot of reduced state transition table constructed manually (not the Quartus auto-generated one)
- Snapshot of RTL post netlist viewer and resource usage table. Explanation of FPGA resource usage in report is not required.

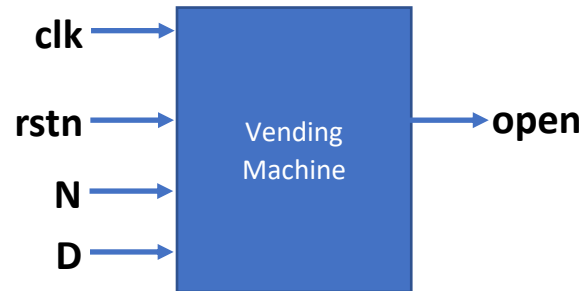
❑ Lab7d folder includes :

- Template design code, full testbench for Vending machine Moore FSM are provided
- Excel file has empty implication chart, reduced FSM transition table. Student can use it to fill in for vending machine. Excel file not required to be submitted. Just take the snapshots from excel file and add in the report !

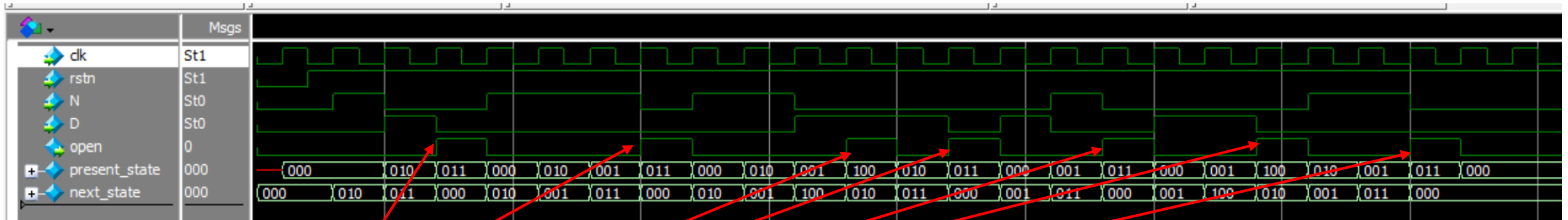
Homework-7d : Vending Machine FSM Minimization

❑ Assume below mentioned primary Ports for Vending Machine

- **Input clk** : posedge clock
- **Input rstn** : reset should be synchronous negedge reset
- **Input N, D** : 1-bit Nickel and Dime inputs indicating Nickel and Dime are deposited if values are '1'
- **Output open** : 1-bit open signal indicating vending machine is dispatching candy



Reference Moore FSM Simulation Waveform



Output "open" is asserted once 15 cents or more are deposited